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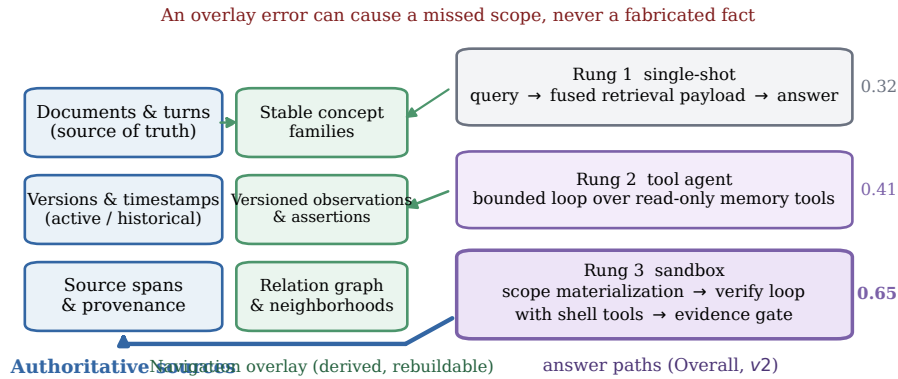


Figure 1: Deep-Dream separates navigation from factual authority. Derived concept families, observations, assertions, and graph neighborhoods form a non-authoritative overlay for scope materialization and candidate generation; the three answer paths differ in how far the answerer may go toward the verbatim sources—fused retrieval payload (rung 1), read-only memory tools (rung 2), or a sandbox of verbatim sources with verification tools and an evidence gate (rung 3). An overlay error can cause a missed scope, never a fabricated fact.

DEEP-DREAM: SOURCE-GROUNDED LONG-TERM MEMORY WITH AUDITABLE CONCEPT OVERLAYS

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ABSTRACT

Long-horizon memory for language-model agents derives compact indexes—facts, summaries, graph edges—whose contribution to final answer quality is rarely separated from the answer path’s access to raw sources. We decompose the two. Deep-Dream is a document-first memory server that demotes all derived structure to non-authoritative navigation: concept overlays materialize *which* sources are in scope, a sandbox exposes the verbatim sources to a verification agent, and an evidence gate requires answers to cite spans the agent actually read, so an overlay error can cause a missed scope but never a fabricated fact. On a 767-question sampled subset of MemoryAgentBench under the official scorer, holding engine, model, and frozen libraries fixed, extending the answer path from single-shot retrieval through memory-tool agency to sandbox-verified reading raises overall accuracy monotonically from 0.324 to 0.651, and the same ladder replicates across two independent engine generations (0.303/0.376/0.669). A channel ablation bounds the derived overlay’s offline retrieval contribution at zero within overlapping confidence intervals, locating quality in the evidence path; the overlay is retained as non-authoritative navigation, with ingest cost falling 46–65% in calls per document between engine generations and consolidation quality maintained (its online selection value not yet separately priced). On the scorer’s judge-free domains the system exceeds published reference points by wide margins, with all caliber differences disclosed.

1 INTRODUCTION

An agent’s memory fails in two ways: it cannot find the past, or it “finds” a past that was never said. Memory systems answer with ever-richer derived indexes—compressed facts, summaries,

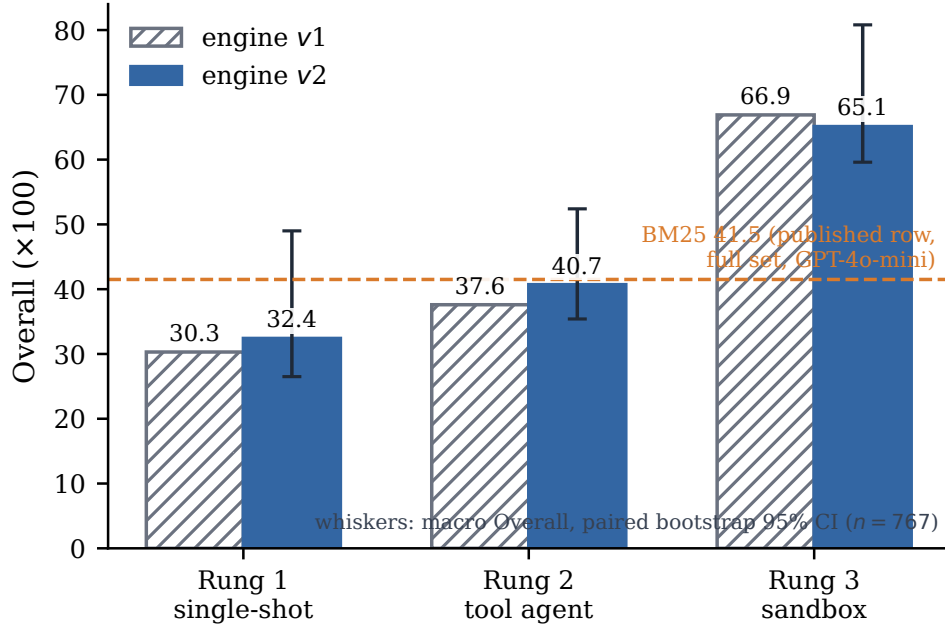


Figure 2: The answer-path ladder on MemoryAgentBench (official scorer, $n=767$, kimi-k3 answerer, frozen libraries per generation): Overall by track for engine $v1$ (hatched) and $v2$ (solid); whiskers, shown on the $v2$ bars only, are paired-bootstrap 95% CIs over the 767 shared questions. The BM25 row (41.5, official paper, full set, GPT-4o-mini) is drawn as a published-reference band, not a same-protocol bar: answerer, judge, and subset differ; no subtraction is implied.

temporal graph edges—and report end-to-end accuracy, but an end-to-end number cannot say where the quality came from: a better index and a better answer path arrive together. This paper asks the question directly: *in long-horizon agent memory, does answer quality come from the derived memory index or from the answer path’s access to raw sources—and if the latter, can derived memory be demoted to non-authoritative, auditable navigation without paying for it in retrieval recall or benchmark overall?* The two answers imply opposite architectures: if quality lives in the index, systems should let derived records authorize answers; if it lives in the evidence path, derived memory should navigate—which sources, which versions—while verbatim sources authorize every answer. Prior systems (Mem0, MemGPT, A-MEM, Zep/Graphiti, HippoRAG-2, Cognee) improve index structure and report accuracy gains, but the two contributions are confounded: no prior comparison holds the memory fixed and varies only the answer path.

Deep-Dream runs that comparison on itself. It is a document-first memory server whose derived layer—stable concept families, versioned observations and assertions, relation edges, converged by cluster alignment—is used for navigation only. The winning answer path works in three moves: the overlay *materializes a scope* (which sources, which versions, are relevant); the scope is exposed as a sandbox of verbatim source documents; an agent verifies and answers inside the sandbox with ordinary tools (file reading, `grep`) and must submit evidence spans it actually read. The submission contract bounds the overlay’s failure mode: an overlay error can cause a *missed scope*, never a fabricated fact (Figure 1).

We evaluate three answer paths on identical frozen libraries—single-shot fused retrieval, a read-only memory-tool agent, and the source-sandbox verification agent—and contribute:

- **An answer-path decomposition.** On a 767-question sampled subset of MemoryAgentBench under the official scorer, accuracy rises monotonically with source access—0.324/0.407/0.651—and replicates across two independent engine generations (0.303/0.376/0.669; $n=767$, answerer kimi-k3, scorer commit 455306d). Because rungs differ as systems, the ladder upper-bounds the marginal effect of source verification; the single-factor evidence is the ablation below.
- **A measured division of labor between index and evidence path.** A channel ablation bounds the derived overlay’s offline retrieval contribution at zero (all channel effects within

overlapping 95% confidence intervals), while the overlay’s measured roles are scope materialization, versioning, and deduplication—navigation, not retrieval. Ingest cost falls 46–65% in calls per document between engine generations with consolidation quality maintained.

- **Cross-system positioning with full caliber disclosure.** On the scorer’s judge-free domains the system exceeds published reference points on length-matched cells (SH-QA 92 vs. best published 87; DetectiveQA 83.3 vs. 77.5), with the larger EventQA and FactConsolidation margins disclosed as length-mismatched (Appendix A.2); subset, answerer, and judge differences are disclosed throughout and no leaderboard subtraction is performed.

Sections 3 to 5 develop the memory model, the decomposition, and what the evidence does and does not prove.

2 RELATED WORK

Long-term conversational memory benchmarks. LoCoMo spans very long conversations with single-hop, multi-hop, temporal, open-domain, and adversarial questions (Maharana et al., 2024); LongMemEval adds multi-session reasoning, knowledge updates, and abstention over 500 questions (Wu et al., 2025); MEME probes evolving entity state, including deletion, cascade, and absence tasks that static retrieval metrics miss (Jung et al., 2026). MemoryAgentBench frames memory as incremental multi-turn interaction across retrieval, test-time learning, long-range understanding, and selective forgetting (Hu et al., 2025), and BEAM extends to conversations of millions of tokens (Tavakoli et al., 2026). These benchmarks are complementary, not interchangeable; we label question count, subset, metric, answerer, and judge on every result.

Structured memory systems. MemGPT established persistent-memory and context-management baselines (Packer et al., 2023); A-MEM organizes experience as linked atomic notes (Xu et al., 2025); Mem0 industrializes incremental memory operations with an LLM judge (Chhikara et al., 2025). HippoRAG-2 couples a knowledge-graph index with hippocampus-inspired retrieval (Gutiérrez et al., 2025), Cognee builds graph memory through extract–consolidate–load pipelines (Markovic et al., 2025), and MIRIX coordinates specialized memory types behind a multi-agent coordinator (Wang & Chen, 2025). All make derived memory a first-class object because compression improves cost and retrieval. We share the need for compact navigation but draw a stronger authority line: a derived node suggests where to look; the source episode remains what the query path must expose and submit.

Long context and retrieval. Retrieval-augmented generation answers from selected sources (Lewis et al., 2020); long-context models keep sources in window but degrade when relevant spans sit far from the window’s ends (Liu et al., 2024). For memory agents this is an architecture axis, not a model property: the official MemoryAgentBench tables show every full-context family failing multi-hop fact consolidation at long range while the same model class solves it at 6K context (Hu et al., 2025). Our rungs deliberately span this axis, and the ladder of Section 4.2 measures where factual authority can sit without paying recall.

Temporal and graph memory. Zep’s Graphiti represents entities and relationships in a temporal knowledge graph (Rasmussen et al., 2025). Deep-Dream also stores relation assertions, timestamps, and concept neighborhoods, but treats the graph as an auditable overlay over documents with an active/incomplete publication boundary: a candidate can be checked against a current source span while historical observations remain addressable.

Evidence and provenance. Attribution research evaluates whether generated text carries citations that support its claims (Gao et al., 2023), but the check is post hoc. Deep-Dream makes it a runtime contract through scope-isolated tools, fresh context per question, read-only access, and evidence-submission validation; the contract is narrower than a semantic fact verifier—it checks reachability, not entailment. Our study asks where the boundary between a useful navigation index and an authoritative source can sit, and evaluates it with end-task QA and retrieval/evidence diagnostics.

3 DEEP-DREAM

3.1 MEMORY MODEL

Let a source document or conversation be D and let e_i denote an episode created from a heading-aware span of D . Deep-Dream stores D , document versions, and episodes as authoritative records. An extracted entity is represented by a stable family identifier f and observations $o_{f,i}$, each linked to an episode and source span; relations use the analogous family/assertion representation. The family is a stable handle for search and alignment; an observation records what a particular source asserted at a particular time, so updating a derived record never overwrites source text. A publication boundary separates states: *complete* documents are searchable while *processing* documents are hidden, and *active* observations answer current retrieval while historical ones remain addressable through provenance, so queries never see partially ingested state.

3.2 REMEMBER PIPELINE

The Remember pipeline chains small, auditable stages—heading-aware chunking; entity and relation discovery with validation and content generation; conservative alignment to existing families; window-batch alignment; cluster convergence—followed by atomic publication. Convergence merges duplicate concept families across ingestion windows: in the paired engine comparison (Section 4.4) it cuts same-name duplicate families from 340 to 124 on MemoryAgentBench scopes and from 482 to 99 on LongMemEval while cutting ingest calls per document 46–65%. Read information-theoretically (Appendix A.13), alignment is a deferred decision: observations are append-only, so early splits are revisable, and a true merge’s detection probability grows exponentially with pooled evidence. Extraction calls may propose entities and relations, but the storage layer records the source episode and provenance rather than treating generated prose as an independent document.

3.3 ANSWER PATHS ON A FROZEN LIBRARY

The paper’s central comparison varies the answer path while holding the memory fixed. All three paths run against the same frozen libraries (identical ingested documents, engine commit, and scope contents), the same question stream, and the same scorer; they differ in how the answer path may reach sources.

Rung 1: single-shot retrieval. One fused retrieval call (lexical, semantic, and graph channels, reciprocal-rank fusion, top- k source spans with neighbor expansion) produces an evidence payload; a separate one-shot answerer consumes question plus payload. No tools, no loop.

Rung 2: memory-tool agent. An agent interacts with read-only memory tools—scope search, entity/relation fetch, provenance lookup—under a bounded step policy, then submits an evidence set through the submission contract; a separate answerer consumes question plus submitted evidence. The agent navigates the derived overlay but never reads raw documents end-to-end.

Rung 3: source-sandbox verification. The agent calls scope materialization, which uses the overlay to select *which* sources (and which versions) are relevant and exposes them as a sandbox of verbatim documents. The agent then verifies and answers inside the sandbox with ordinary tools—file reading, `grep`, small scripts—and outputs the answer together with the evidence spans it read.

The rungs differ as *systems* (Table 1): tools, answerer arrangement, loop, prompt, and compute change together, so the ladder upper-bounds the marginal effect of source verification; the single-factor evidence is the channel and gate ablations (Section 4.5).

The overlay’s role in the winning path deserves emphasis: the ablation shows its retrieval channels are recall-null. In rung 3 the overlay decides *which sources are in scope*, the sandbox holds verbatim sources, and the gate turns any overlay error into a missing file—a missed scope—never a fact the agent could still assert. Its measured roles are scope materialization, versioning, and deduplication: navigation, not retrieval.

	Rung 1 single-shot	Rung 2 agent	tool	Rung 3 sandbox
Source access	top- k spans from one fused call	derived records via tools		whole sources in sandbox
Tools	none	read-only memory tools		shell (grep, read, script)
Answerer	separate, one-shot	separate, submitted evidence	reads evi-	the verifying agent itself
Loop	none	bounded loop	tool	bounded verification loop
Prompt	QA over payload	tool policy + submit		verify-then-answer policy
Compute / q	25.1k tokens, 1.0 LLM calls	35.0k tokens, 6.1 LLM calls		96.0k tokens, 6.6 LLM calls

Table 1: What changes at each rung. All rungs share frozen libraries, question stream, answerer model (kimi-k3), and scorer; compute per question is recomputed from run logs (Section 4.2).

3.4 HYBRID NAVIGATION AND SOURCE EXPANSION

For a query, candidate records come from lexical search, semantic similarity, concept search, relation evidence, and graph-neighborhood expansion, combined by reciprocal-rank fusion without requiring one channel to dominate. The query path resolves candidates to source episodes and spans and expands to neighboring turns when an exact span is too narrow for a multi-turn question. This is a retrieval policy, not a claim that more context is always better; the experiments report evidence recall together with response size.

3.5 EVIDENCE GATE

Each benchmark question runs in a fresh conversation scope: the agent cannot see the gold answer, gold evidence identifiers, or other conversations, and its tools are read-only. The final submission must satisfy:

1. every submitted identifier was returned by a tool call in the current question;
2. the identifier belongs to an active, eligible scope and document state;
3. source-level identifiers are retained so an independent auditor can inspect the exact text;
4. the answer and trajectory are stored separately from hidden reasoning text.

The contract enforces provenance reachability, not semantic entailment—a model can still submit a relevant-looking source and produce an unsupported answer. The gate is verified deterministically, without any LLM, in Section 4.5, together with deployed-run rejection statistics.

3.6 IMPLEMENTATION AND AUDIT ARTIFACTS

The runtime records dataset and prompt provenance, model identifiers, tool observations, evidence IDs, latency, exit status, and code/runtime hashes for every question. The machine-readable evidence ledger is generated from frozen JSON summaries and per-item JSONL files; failed integrity checks and negative results are preserved in the paper input rather than dropped.

4 EXPERIMENTS

4.1 SETUP AND CALIBER DISCIPLINE

MemoryAgentBench (MAB) (Hu et al., 2025) is the paper’s primary instrument because its official scorer and published per-domain tables permit a cross-system comparison under one protocol. We run a stratified sampled subset: 767 questions over 10 scopes and 59 documents, against the benchmark’s

Answer path ($n=767$)	Overall		Sandbox per-domain	
	$v1$	$v2$	domain	$v1 \rightarrow v2$
Rung 1 single-shot	30.3	32.4	AR (SH-QA / LME-S* / EventQA)	94.0 $\rightarrow$ 92.8
Rung 2 memory-tool agent	37.6	40.7	TTL (MCC, icl_clinic150)	43.0 $\rightarrow$ 46.0
Rung 3 source-sandbox	66.9*	65.1	LRU (∞ Bench-Sum / DetQA)	52.3 $\rightarrow$ 41.7
			SF (FC-SH / FC-MH)	78.5 $\rightarrow$ 80.0

Table 2: The answer-path ladder, official MemoryAgentBench scorer, kimi-k3 answerer, identical frozen libraries and question stream per generation; per-domain cells are shown for the sandbox rung (rungs 1–2 in Appendix A.4). *The $v1$ Overall carries the $n=1$ ∞ Bench-Sum flip (0.212 $\rightarrow$ 0.000); excluding that task $v2$ leads 75.5 vs. 74.7. Rungs differ as systems (Table 1); deltas are consistent-with readings, not single-factor estimates.

full 3,671 questions over 29 tasks. The subset covers all four competencies but not every task (task lists in Appendix A.2); all MAB numbers below carry this subset statement, with a composition-adjusted Overall beside any Overall figure (Section 4.3). Scoring uses the official MemoryAgentBench scorer at commit 455306d with an identical `entity2id` fingerprint across runs; kimi-k3 is the answerer (and the LME-domain judge; the official protocol likewise uses an LLM judge on that domain) for all three tracks, and the ingest model for all libraries. The MAB source corpora are public; we cannot rule out pretraining contact and flag this wherever cross-system numbers lead.

LongMemEval-S (LME) full ingest covers 1,176 documents over 25 scopes with per-scope libraries. Track accuracies use paired same-question sets of $n=25$ (baseline, skill-agent) and $n=27$ (sandbox; two extra questions under its session-evidence eligibility rule), so cross-track comparisons are same-track $v1 \rightarrow v2$ only; we label these small n wherever they appear and treat them as ancillary. The 500-question LME-S run (84.60%, Qwen3.7-plus cross-judge) is reported separately and protocol-labeled; LoCoMo-1540 under Mem0’s exact binary- J protocol and MEME are context and limitations respectively. BigCodeBench (0.4859/0.2838 completion/hard pass@1, calibrated) (Zhuo et al., 2024) and ALFWorld (0.9786/0.9851 id/ood) (Shridhar et al., 2021) are memory-insensitive agent-carrier anchors (appendix).

4.2 THE ANSWER-PATH LADDER

Table 2 and Figure 2 give the headline result. Holding engine, model, frozen libraries, question stream, and scorer fixed, accuracy rises monotonically with source access at every rung— $v2$ Overall 0.3240/0.4068/0.6511, $v1$ 0.3025/0.3759/0.6695—and the ladder replicates across two independent engine generations ($n=767$ per rung per generation). Paired bootstrap over the 767 shared questions puts the rung deltas at +0.083 [+0.015, +0.149] (rung 2 – rung 1) and +0.244 [+0.214, +0.305] (rung 3 – rung 2). Per-rung compute, recomputed from run logs, is 25.1k tokens, 1.0 LLM calls, 35.0k tokens, 6.1 LLM calls, and 96.0k tokens, 6.6 LLM calls per question (Figure 3): the top rung buys its accuracy with an agentic verification loop.

Two disclosures bound the reading. First, the rungs differ as *systems* (Table 1), so the ladder’s deltas are consistent with source verification carrying quality and upper-bound its marginal effect—not a single-factor estimate; the single-factor evidence appears in Section 4.5. Second, rung 1 is not a strawman: the official BM25 row (simple RAG, GPT-4o-mini, full set) scores 41.5 Overall—above our rung 1’s 32.4, a disclosed cross-model, cross-subset difference—and collapses on FactConsolidation-MH (3.0, at 262K-average contexts) while our rung 1 collapses on the 6K variant (2–4); a single retrieval shot does not crack consolidation at either length, the verification loop does.

The LME full-ingest tracks echo the ordering at different magnitudes: 0.72/0.72/0.926 (baseline/skill/sandbox, $n=25/25/27$, $v2$ engine)—skill and baseline tie in both generations while the sandbox path, on its paired set with a differing eligibility rule (Section 4.1), scores 0.926 in $v2$ and 0.889 in $v1$, consistent with the evidence path carrying the quality.

4.3 CROSS-SYSTEM POSITIONING UNDER THE OFFICIAL SCORER

Table 3 places the sandbox track (top rung; lower rungs in Table 2) beside the official paper’s Table 2 rows, which we cite verbatim rather than re-run (Hu et al., 2025). On the scorer’s judge-free domains—substring match or fixed accuracy, immune to judge-model identity—the sandbox track exceeds

every published row on the length-matched cells (SH-QA 92 vs. best published 87; DetectiveQA 83.3 vs. 77.5) and posts larger raw margins on EventQA (98 vs. 82.6; sampled at 64K—full-book scopes against their 534K full-book average) and FactConsolidation (SH 90 vs. 60; MH 70 vs. best published 7; our 6K variants against their 262K-average contexts)—mismatched-length cells we read as directional, with the FC accounting below and full scope lengths in Appendix A.2. The composition-adjusted Overall is 59–61 against a best published 49.6 (derivation in Appendix A.3); we never subtract across calibers—rows differ in answerer, judge, and subset, and the table states each.

The model-substitution question. Our rows use kimi-k3 where published rows use OpenAI, Google, and Anthropic models; a skeptic can ask whether a stronger model, not the architecture, explains the margin. No same-model full-context anchor exists (the kimi-k3 endpoint was retired mid-project; all results are frozen per-item artifacts)—stated as a limitation. Four published facts bound the alternative explanation. (i) Five full-context models across three families—GPT-4o (48.8), GPT-4o-mini (42.2), GPT-4.1-mini (46.9), Gemini-2.0-Flash (42.4), Claude-3.7-Sonnet (49.6)—all fail FactConsolidation-MH (≤ 5 , at 262K-average contexts) and none exceeds 49.6 Overall. (ii) The official backbone ablation (their Table 3) finds that once a backbone is sufficiently strong it is not the bottleneck for retrieval-path agents. (iii) The profile is asymmetric the way an architecture effect predicts and a model effect does not: we *trail* the best published full-context row on the length-matched TTL (46 vs. 53.9) and LRU (41.7 vs. 62.2) yet lead on AR (92.8 vs. 71.8; SH-QA 92 vs. 87 is the cleanest matched cell) exactly along the benchmark’s own taxonomy; task-matched imputations for the omitted tasks (AR ≈ 84.6 , TTL ≈ 29.5 , Appendix A.3) leave the asymmetry intact under uniform model inflation. (iv) The SF leg needs length accounting: the official length-validation (their Table 4) shows o4-mini at 80 on the 6K FactConsolidation-MH variant but 14 at 32K—published FC failure is long-context interference, not capability—so the raw FC juxtaposition above is inflated by the mismatch in our favor, and we claim only the rung separation on identical scopes. Cross-model asymmetry is a property of the official table’s construction—its memory rows run GPT-4o-mini against frontier full-context models—not something our row introduces.

As protocol-matched context, LoCoMo-1540 under Mem0’s exact binary- J protocol with kimi-k3 as both answerer and judge scores 95.19 (1,466/1,540, single run), 94.22 under an independent Qwen3.7-plus cross-judge on the same answers, 98.51% cross-judge agreement (Table 6 in Appendix A.12). The two admitted differences from Mem0’s published J are a single run versus their 10-run mean $\pm$ std and the model substitution, and the run carries a disclosed legacy-fingerprint caveat—reported as protocol-matched context, never a superiority claim.

4.4 PAIRED ENGINE COMPARISON: WHAT NAVIGATION COSTS AND KEEPS

The $v1 \rightarrow v2$ engine upgrade (cluster convergence atop window-batch alignment) is evaluated as paired runs on identical inputs. Ingest efficiency: LME calls/doc 63.8 $\rightarrow$ 22.0 (−65.5%), tokens/doc 139.0k $\rightarrow$ 111.7k (−19.6%); MAB calls/doc 609 $\rightarrow$ 329 (−46%), tokens −18% (45-document paired intersection; manifest gaps in Appendix A.7). Consolidation quality is maintained: the two target domains move +0.03 (TTL MCC 0.43 $\rightarrow$ 0.46, FC-MH 0.67 $\rightarrow$ 0.70; paired CIs +0.030 [−0.070, +0.130] and +0.030 [−0.070, +0.130])—within-sample noise we do not interpret as gains; same-name duplicate families fall from 340 to 124 on MAB scopes and from 482 to 99 on LME (recomputed from the per-scope library databases, Appendix A.1). The $v2$ Overall on MAB (0.6511) sits 0.018 below $v1$ (0.6695); the gap is dominated by the $n=1 \infty$ Bench-Sum flip (0.212 $\rightarrow$ 0.000; excluding it $v2$ leads 0.008). Retrieval on the LME baseline track rises recall@10 0.767 $\rightarrow$ 0.847 and nDCG@1 0.720 $\rightarrow$ 0.800.

4.5 RETRIEVAL DIAGNOSTICS: CHANNEL EQUIVALENCE BOUND, GATE, BUDGET

Channel ablation (X7, offline). The derived overlay’s retrieval contribution is measured on the 210-question development/gate set with frozen libraries and one factor changed per arm: source-only lexical, +semantic, +graph-neighbor, +relation (full five channels). Recall-any@10 is 72.38/72.38/71.43/72.38% across arms—every channel effect within ± 2.4 points at 95% (percentile bootstrap, 2,000 resamples, CIs fully overlapping); recomputed recall@1, MRR, and nDCG@10 agree (recall@1 25.7/24.8/25.7/26.2, MRR@10 40.2/40.1/40.5/41.0, nDCG@10 42.0/42.2/42.2/42.7; per-arm CIs and payload sizes in Table 5, appendix). The honest reading is an

	System (answerer)	AR	TTL	LRU	SF	Overall
ours [†]	Rung 3 sandbox (kimi-k3)	92.8	46.0	41.7	80.0	65.1
full-ctx	GPT-4o	58.1	50.0	54.9	32.5	48.8
	GPT-4o-mini	49.2	48.6	46.2	25.0	42.2
	GPT-4.1-mini	71.8	46.2	49.1	20.5	46.9
	Gemini-2.0-Flash	65.1	46.4	41.6	16.5	42.4
	Claude-3.7-Sonnet	59.7	53.9	62.2	22.5	49.6
published	BM25 simple RAG (GPT-4o-mini)	60.5	44.5	35.6	25.5	41.5
	HippoRAG-2 (GPT-4o-mini)	65.1	35.8	36.2	29.5	41.6
	MIRIX (GPT-4.1-mini)	63.0	35.7	40.5	11.5	37.7
	MemGPT (GPT-4o-mini)	34.3	40.8	22.4	15.5	28.3
	Zep (GPT-4o-mini)	37.5	37.5	16.2	5.0	24.0
	Mem0 (GPT-4o-mini)	32.6	21.2	20.7	10.0	21.1

Table 3: Cross-system positioning on MemoryAgentBench under the official scorer (scores $\times 100$; AR/TTL/LRU/SF are domain averages). [†]Our row uses the stratified sampled subset (767 questions, 10 scopes, 59 documents; all four domains, 8 of 29 tasks), kimi-k3 answerer, official scorer at commit 455306d; the rung 1/2 Overall values (32.4/40.7) appear in Table 2, and the adjusted Overall (59–61) in Appendix A.3. Published rows are cited verbatim from the official paper’s Table 2 (Hu et al., 2025) (full benchmark; answerers as listed; memory-system rows run GPT-4o-mini except MIRIX on GPT-4.1-mini). Every full-context family scores at most 5 on FactConsolidation-MH; best published FC-MH of any system is 7 (published FC columns at 262K-average contexts; our SF cells are the 6K variants, Appendix A.2). No two rows share answerer, judge, and subset; we never subtract across calibers.

equivalence bound: on this slice the derived channels do not add offline retrieval recall, and the null speaks only to offline retrieval—in the sandbox path the overlay’s online role (scope materialization, versioning, deduplication) is *exercised*, not ablated.

Evidence gate. Enforcement is proved deterministically, without any LLM: an identifier that exists in the scope but was never surfaced by a read tool is rejected with the gate ON and accepted after the one-flag relaxation. Deployed runs: 11/767 questions in the deployed *v2* skill-agent run had an evidence submission rejected as unsurfaced (9/767 in *v1*); 9 of the 11 resubmitted tool-surfaced evidence inside the loop and the remaining 2 ended at the step cap and were answered from tool-surfaced context; zero rejections on the LME runs (Table 5, appendix).

Budget frontier. An earlier depth-10-versus-depth-20 replay is retained as an integrity finding: its prefix invariant failed on all 210 questions. The repaired design retrieves once per query at maximum depth and slices a fixed ranking to $k \in \{1, 3, 5, 10, 20\}$ —prefix monotonicity holds by construction, and the frontier rises from $k=1$ to $k=20$ with bootstrap CIs (Appendix A.8).

4.6 INTEGRITY AND NEGATIVE RESULTS

Negative results, stated plainly. (i) TTL is the structural gap (46 vs. published full-context 53.9–96 on MCC tasks): full-context agents keep labeled in-context examples verbatim while any memory system compresses them, mirroring the official finding that full-context models win TTL and LRU; the lever—denser atomic-fact extraction—is future work. (ii) MEME deletion is 0% (100 episodes, 694 after-task checks)—architectural: the memory model keeps every observation addressable and implements no deletion; negative-memory semantics (cascade 46.34, absence 59.23 real accuracy) remain central limitations. (iii) Identical Qwen3.7 LoCoMo-1540 answers move 92.21/93.18/93.57 under prompt and judge substitutions, and substituting kimi-k3 as answerer under the same protocol gives 95.19 (cross-judge agreement 98.51%), so every table labels answerer, judge, and prompt provenance. (iv) Residual failures: the sandbox track’s macro-averaged Overall of 0.6511 leaves a 34.9% gap in Overall terms, concentrated in 117 zero-score questions of 767 (15.3% micro-average); their recomputed taxonomy is 46.2% test-time learning, 34.2% selective forgetting (30/10 FC-MH/FC-SH questions), 17.9% long-context reading, 1.7% long-range understanding, and the largest single LRU contributor is the $n=1$ Summ flip (0.212→0.000), disclosed wherever Overall $v1 \rightarrow v2$ is compared—the full non-sampled benchmark run is the right instrument for that domain and has not been run.

5 ANALYSIS AND LIMITATIONS

5.1 WHAT THE DECOMPOSITION DOES AND DOES NOT SHOW

The ladder of Section 4.2 is a systems comparison (Table 1): each rung delta is *consistent with* source verification carrying quality and upper-bounds its marginal effect, never a single-factor estimate—the single-factor evidence is the channel ablation’s equivalence bound plus the gate’s deterministic enforcement, which locate the overlay’s value in navigation, not recall; replication across engine generations is pattern robustness, not causal isolation. The strongest honest statement: the evidence path carries the quality, and the demoted index did not pay for the demotion in recall or in Overall.

5.2 POSITIONING, READ CONSERVATIVELY

Three reading rules govern Table 3. First, no same-model full-context anchor exists: the kimi-k3 endpoint was retired mid-project and all results are frozen per-item artifacts. Second, our rows are a sampled subset (767 of 3,671); the composition-adjusted Overall is 59–61 (Appendix A.3) against a best published 49.6 beside the raw number. Third, the LME(S*) cell (88.3 vs. 55.7) is the one judge-involved comparison; the headline cells (SH-QA 92 vs. 87, DetectiveQA 83.3 vs. 77.5, EventQA 98 vs. 82.6, FC-MH 70 vs. 7) are substring/fixed-accuracy scored and immune to judge concern, though EventQA and the FC cells carry disclosed length mismatches (Appendix A.2).

5.3 LIMITATIONS

(1) No same-model anchor; the model-substitution objection is bounded by four published facts (Section 4.3), not closed. (2) Sampled subset (8 of 29 tasks over 10 scopes); omitted low-scoring tasks bias the raw Overall *up*, corrected by the adjusted estimate. (3) LME full-ingest uses $n=25/27$ paired samples—labeled, small, ancillary. (4) The sandbox path costs an agentic loop (96.0k tokens, 6.6 LLM calls vs. 25.1k tokens, 1.0 LLM calls per question), counted as part of the result. (5) MEME deletion is 0%; negative-memory semantics need a query-time suppression filter (future work). (6) TTL trails full-context structurally; the lever (atomic-fact extraction density) is untested. (7) LoCoMo $J=95.19$ is a single run with a disclosed legacy-fingerprint caveat (Appendix A.12). (8) MAB corpora are public; pretraining contact cannot be excluded post hoc. (9) The LME skill–baseline tie is a property of that paired set.

5.4 FUTURE WORK

The negatives name the next experiments: a query-time suppression filter for deletion and absence semantics; denser atomic-fact extraction for TTL; a trivial-selector control (BM25/embedding top- m scope materialization) to price the overlay’s navigation role; the full 3,671-question run; and a fingerprint-complete same-model rerun when a stable endpoint exists.

6 CONCLUSION

This paper asked where answer quality lives in long-horizon agent memory—in the derived index or in the answer path’s access to raw sources—and built Deep-Dream to run the comparison. The answer the data supports: in the evidence path. Extending the answer path from single-shot retrieval through memory-tool agency to sandbox-verified reading raises official-scorer accuracy monotonically from 0.324 to 0.651 and replicates across two engine generations; the overlay’s retrieval channels are recall-null; its online selection value is not yet separately priced, so it stays as non-authoritative navigation at 46–65% lower ingest calls per document. Because the evidence gate requires answers to cite spans the agent actually read, an overlay error can cause a missed scope but never a fabricated fact.

We do not claim single-factor causal isolation, a same-model full-context anchor, a full-set run, or solved deletion and test-time learning. But the ladder’s replication and agreement with the ablations make one lesson hard to avoid—let verbatim sources authorize answers, and let derived memory only decide where to look.

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A REPRODUCIBILITY AND AUDIT DETAILS

A.1 ARTIFACT LEDGER

The machine-readable ledger is generated by `paper/results/aggregate_existing.py`. It records the source summary, dataset, and per-item JSONL SHA-256 values for every run the paper cites, and is regenerated and compared byte-for-byte with the script output before paper writing. The current ledger covers the MemoryAgentBench tracks (two engine generations $\times$ three rungs, official scorer at commit 455306d, identical `entity2id` fingerprint), LongMemEval full-ingest pairs and the 500-question run, LoCoMo-1986/1540 and the K3-as-answerer diagnostic, LoCoMo-Plus-401, the X7 channel diagnostic, the source-expansion diagnostic, the repaired budget-frontier replay, MEME, BigCodeBench, and ALFWorld. Per-run manifests record dataset and prompt provenance, model identifiers, tool observations, evidence IDs, latency, exit status, and code/runtime hashes; failed integrity checks and negative results are retained rather than dropped. The ledger-generation and recompute scripts ship with the released supplementary material so the chain is regenerable end to end.

The main local artifacts are:

- MemoryAgentBench tracks: `.benchmark_runs/<run>/memoryagentbench_summary.<track>...sampled.json` with per-item `memoryagentbench_scores.<track>...jsonl`; sampling frame from `mab_sample.py` (`mab-sample.scopes.txt`, `.stats.json`).
- LoCoMo-1540 exact judge summary: `.benchmark_runs/locomo-full-quality-v1/judge_summary.kimi-agent-direct-qwen37-full-thinking-off.qwen37-mem0-current-exact-direct.json`; the judge-prompt-substitution point (92.21) is `...qwen37-kimi-full.json` in the same run directory.
- Engine-comparison ingest efficiency (Section 4.4) derives from per-document `llm_call_stats` in the run manifests of the same MAB/LME run directories; the ledger’s `engine_comparison` block recomputes the per-document ratios from the hashed manifests (LME 940/1020 docs with stats; MAB 58/46 with stats, 45-document paired intersection) and the library-level counts (document totals, entity-family totals, duplicate-name statistics over normalized `canonical\_name`) from the hashed per-scope library databases.
- K3-as-answerer diagnostic: `.../judge_summary...legacy-fingerprint-diagnostic.json` under `.benchmark_runs/locomo-k3-agent-judge-diagnostic-v1/` (primary, 95.19%) and the independent Qwen3.7-plus cross-judge (94.22%).
- LongMemEval-S-500 summary: `.benchmark_runs/longmemeval-source-v24-full500/judge_summary...memory-v24-full500.json`.
- LoCoMo-Plus-401 summary: `.benchmark_runs/locomo-plus-cognitive-qwen35-cues-full-v1/judge_summary...full-v1-final2.json`.

A.2 MEMORYAGENTBENCH SAMPLING FRAME AND CONTAMINATION NOTE

The sampled subset was drawn by `mab_sample.py` with a fixed scope list recorded in `mab-sample.scopes.txt`: 767 questions over 10 scopes and 59 documents against the full 3,671 questions over 29 tasks. All four competencies are covered; the omitted tasks are AR’s MH-QA, TTL’s Recommendation, and 19 tasks whose scopes were not sampled. The same subset and `entity2id` fingerprint are used for all six ladder runs (three rungs $\times$ two generations). Scope lengths are heterogeneous and disclosed wherever cross-system cells lead: the two FactConsolidation scopes are the 6K variants (26,157 characters each); the 32K/64K/262K FC variants (136k–1.1M characters), the 421K RULER scope, and ten other scopes (17 dropped in total) were dropped for budget, all listed with context sizes in the released ledger’s subset block; EventQA is sampled at the 64K/128K/full-book scopes rather than only the published full-book average; and the LRU domain

carries $n=7$ questions (∞ Bench-Sum $n=1$, DetectiveQA $n=6$), so LRU cells are directional. The MAB source corpora are public; whether the answerer’s pretraining saw them cannot be excluded or measured post hoc, and every cross-system table row states its own answerer, judge, and subset.

A.3 ADJUSTED OVERALL DERIVATION

The raw Overall (65.1) averages our four domain scores, but two sampled domains omit tasks that score low for everyone. Imputing the omitted tasks at published non-full-context reference values—TTL: our MCC 46 with Recommendation ≈ 13 gives a TTL average ≈ 29.5 ; AR: our three tasks (92 / 88.3 / 98) with MH-QA ≈ 60 gives an AR average ≈ 84.6 ; LRU 41.7 and SF 80.0 unchanged—yields an adjusted Overall of ≈ 59.0 , and varying the imputed values across the published non-FC range gives 59–61. We report the adjusted estimate beside the raw one because the raw one is biased upward by composition; both are stated beside the best published full-context row (49.6), which differs in answerer, judge, and subset—the comparison is contextual, and no claim in the paper subtracts across calibers.

A.4 MAB PER-TASK DETAIL

Task (domain)	Baseline		Tool agent		Sandbox	
	$v1$	$v2$	$v1$	$v2$	$v1$	$v2$
SH-QA (AR)	54	48	84	72	94	92
LongMemEval-S* (AR)	26.7	41.7	43.3	51.7	90.0	88.3
EventQA (AR)	90.3	89.7	90.7	88.3	98.0	98.0
MCC icl_clinic150 (TTL)	4	2	3	2	43	46
∞ Bench-Sum (LRU)	0.0	0.0	28.4	29.8	21.2	0.0
DetectiveQA (LRU)	50.0	66.7	50.0	83.3	83.3	83.3
FC-SH (SF)	68	66	67	63	90	90
FC-MH (SF)	2	3	4	4	67	70

Table 4: Per-task scores ($\times 100$) under the official scorer for all six ladder runs (three rungs $\times$ two engine generations); every cell is read from the released evidence ledger (Appendix A.1), and domain aggregates and per-rung compute appear in Table 2. ∞ Bench-Sum has $n=1$ in the sampled subset—the 21.2 $\rightarrow$ 0.0 sandbox flip is a single-question event, disclosed wherever Overall $v1\rightarrow v2$ is compared. The consolidation tasks separate the rungs most sharply: single retrieval and the tool rung stay at or near zero at both generations, and only the sandbox rung lifts them (MCC 43–46, FC-MH 67–70).

A.5 PER-RUNG PROMPTS AND CONFIGURATIONS

All rungs use kimi-k3 as answerer with per-rung prompts archived verbatim in the run manifests and code snapshots listed in Appendix A.1; the released supplementary material reproduces them in full. The $v1\rightarrow v2$ engine configurations differ as: max output tokens 16,384 $\rightarrow$ 32,768; usable context window 32k $\rightarrow$ 262,144; retries 2 $\rightarrow$ 5; timeout 300 s $\rightarrow$ 1,200 s; absolute-path persistence enabled in $v2$. Rung policies: rung 1 one fused retrieval call (top- k spans, neighbor expansion) plus one answer call; rung 2 bounded tool loop over read-only memory tools with a separate answerer over submitted evidence; rung 3 scope materialization then a bounded verify-then-answer loop with shell tools and evidence-gated submission.

A.6 CARRIER ANCHORS (BIGCODEBENCH, ALFWORLD)

BigCodeBench: 0.4859 completion / 0.2838 hard pass@1 (calibrated split). ALFWorld: 0.9786 in-distribution / 0.9851 out-of-distribution success. Both are memory-insensitive agent-carrier regressions: they anchor that the agent substrate executes tasks correctly, and are not compared horizontally with memory benchmarks.

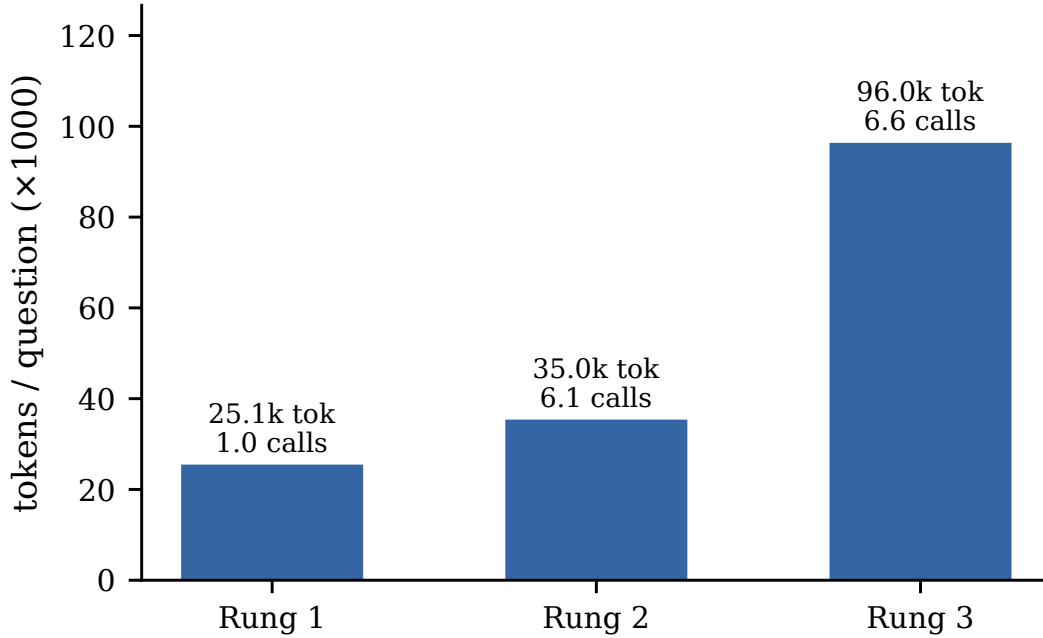


Figure 3: Per-question compute by rung, recomputed deterministically from run logs (tokens and LLM calls, mean over the 767 questions; *v2* engine). The single-shot rung is one LLM call over a 25k-token fused retrieval payload; the top rung buys its accuracy with an agentic verification loop at $3.8\times$ the tokens. The cost accounting is part of the result, not a footnote.

A.7 MAB INGEST MANIFEST GAPS

The *v1* and *v2* MAB ingest runs share a 45-document paired intersection. Manifests missing from one generation are excluded from per-document efficiency ratios, which are therefore computed on the intersection only (609→329 calls/doc, -46%); token ratios use the same intersection. Question-level results are unaffected: all 767 questions run against frozen libraries in both generations.

A.8 BUDGET FRONTIER

An earlier depth-10-versus-depth-20 replay is retained as an integrity finding: its prefix invariant failed on all 210 questions (`primary_result_invariant_violations=210`), so it is not used for any positive claim. The repaired design runs retrieval once per query at maximum depth and slices a fixed ranking to $k \in \{1, 3, 5, 10, 20\}$; prefix monotonicity then holds by construction (zero violations), and the frontier rises monotonically from $k=1$ to $k=20$ with per- k percentile-bootstrap CIs recomputed from the frozen replay artifacts.

A.9 X7 CHANNEL ABLATION DETAIL

A.10 PROTOCOL BOUNDARIES

The original LoCoMo token-F1 track includes 1,986 questions and five categories. The Mem0-compatible track contains 1,540 questions and four categories. LongMemEval-S contains 500 questions; the full-ingest paired tracks use $n=25/25/27$ as described in Section 4.1. MEME’s 100 episodes produce heterogeneous task counts (ER/Agg/Tr/Del/Cas/Abs = 100/100/100/100/164/130 after-task checks in the cited run). Raw and real scores are not interchangeable.

A.11 INTEGRITY FAILURES RETAINED

The depth replay of Appendix A.8 reports `primary_result_invariant_violations=210` and `passed=false`. The MEME deletion result is 0% real accuracy. The X7 channel ablation is a null result, reported as an equivalence bound rather than dropped. The ∞ Bench-Sum $n=1$ flip

Table 5: X7 provenance ablation, offline retrieval replay on the 210-question gate set. Arms 1–4 form a clean channel ablation: only `enabled_channels` differs, so any recall delta is attributable to that channel set. Arms 5 (evidence-bound agent, gate ON) and 6 (unsurfaced evidence, gate relaxed) reuse arm 4’s five-channel ranking and are retrieval-identical to arm 4; their distinguishing columns (QA accuracy, unsupported-answer rate, mean evidence tokens) require the LLM agent run and are pending. The evidence-gate plumbing was proved offline: with the gate ON, submitting an unsurfaced turn is rejected; with the gate relaxed the same submission is accepted (`plumbing_ok=true`). In the deployed MAB runs the gate fired as designed: 11/767 questions in the deployed *v2* skill-agent run had an evidence submission rejected as unsurfaced (9/767 in *v1*); 9 of the 11 resubmitted tool-surfaced evidence inside the loop and the remaining 2 ended at the step cap and were answered from tool-surfaced context; zero rejections on the LME runs. Bootstrap CIs: 2,000 resamples, 95%.

Arm	Channels (additive)	Recall-any@10 [95% CI]	Recall-all@10 [95% CI]	Mean bytes@10
1	raw-doc + episode-bm25	72.38 [66.67, 78.57]	51.43 [44.76, 58.10]	1833
2	+ semantic-provenance	72.38 [66.67, 78.57]	51.43 [44.76, 58.10]	1841
3	+ graph-neighbor	71.43 [65.70, 77.62]	50.95 [44.29, 57.62]	1845
4	+ relation-evidence (full 5)	72.38 [66.65, 78.57]	51.90 [45.24, 58.58]	1847

costs the *v2* Overall 2.7 points directly (the net *v1*→*v2* gap is 1.8 points) and is disclosed at every comparison. The LME skill–baseline tie is reported although it is unflattering to the tool-agent rung. These records remain visible because an auditable memory paper should expose failed and null mechanisms as well as successful ones.

A.12 EXTERNAL COMPARISON PROTOCOL

Mem0’s original paper reports LoCoMo category-level F1, BLEU-1, and a binary *J* in which a GPT-4o-mini judge labels each answer CORRECT or WRONG and the score is the percent judged CORRECT (mean±std over 10 independent runs). Zep’s original paper reports DMR and LongMemEval with GPT-4o-family answerers. Rather than re-run those systems under our protocol, we reproduce their published tables verbatim below (Tables 7–9) as cross-protocol contextual references. The Mem0 deterministic token-overlap metrics omitted from the main tables for space are reproduced in Table 11.

The K3-substitution row. The single Deep-Dream *J* row placed inside Mem0’s *J* column is produced by running Mem0’s exact binary-judge protocol with Kimi-K3 substituted for GPT-4o-mini as *both* the answerer and the judge—the only change is the model. This yields 95.19% under the same-model K3 judge (97.38/93.62/80.21/95.33 S/M/O/T) and 94.22% under an independent Qwen3.7-plus cross-judge on the same K3 answers (96.91/92.91/77.08/93.46; cross-judge agreement 98.51%). Two differences from Mem0’s published *J* are admitted: it is a single run rather than a 10-run mean±std, and the model is Kimi-K3 rather than GPT-4o-mini. We additionally disclose an integrity caveat: the run manifest carries `diagnostic_override={kind:legacy-missing-runtime-code-fingerprint, formal_claim_allowed:false}`, because 1,529 of the 1,540 answer rows originate from a legacy source run (`locomot-full-quality-v1`) that predates runtime-code fingerprinting. The answers themselves are unchanged (`answer_rows_unchanged=true`, answer SHA-256 76af5772) and the per-item JSONL carries `runtime_code_sha256` on all 2,063 rows (3 distinct hashes: 2fa32774 on 2,019, 36b45933 on 33, 42afb1ac on 11), the same dataset SHA-256 (79fa87e9) and library snapshot (cf2e76b8) as the Qwen-answerer run, and the same Mem0 binary-judge prompt (repository `mem0ai/memory-benchmarks`, commit 4b61c5d, mode `unified-no-evidence-exact-direct`). We therefore report 95.19% as a diagnostic cross-judged result, not a fingerprint-complete formal entry.

A stricter controlled comparison—freezing the same answer model, judge model, prompt, temperature, evidence budget, dataset revision, and Graphiti/Zep implementation commit—remains the only path to a shared leaderboard, but it is not a submission hard gate now that the published numbers are cited directly.

A.13 AN INFORMATION-THEORETIC READING OF THE ARCHITECTURE

This appendix gives the paper’s two design commitments a formal shape. Every statement is either a structural property of the architecture (checkable against the schema and code) or a conditional proposition with explicit assumptions; none upgrades the causal wording discipline of Sections 4.2 and 4.5.

Write time: alignment as deferred decision. Let the source stream be immutable documents, each extracted mention carry a feature vector (name and content embeddings, attribute assertions, relation-endpoint co-occurrence), and each family hold an append-only observation list linked to exact source spans. Merging is non-destructive—the losing family’s observations re-attach with provenance, and merges are idempotent—so alignment errors are revisable coarsenings, not information loss.

Proposition A.1 (Revisability). *In the append-only data model, any coarsening of the current family partition is realizable at any later time by non-destructive merges, with no loss of evidence. A store that instead overwrites an entity record with a running summary σ satisfies $I(Z; \sigma) \leq I(Z; O)$ (data processing inequality), strictly when the summary discards identity-discriminative detail—a loss no later evidence can restore.*

First-shot alignment accuracy is therefore not the quantity that matters; evidence retention is. The system’s conservative thresholds (low-confidence merges degrade to creating a new family) deliberately bias early errors toward over-splitting—the recoverable direction, since a wrong merge contaminates one handle with two entities’ evidence and is strictly harder for the read path to detect.

Proposition A.2 (Convergence under pooled evidence). *Assume observation features are i.i.d. given entity identity, with distinct entities $e \neq e'$ emitting $P_e \neq P_{e'}$ of Chernoff information $C(P_e, P_{e'}) > 0$. Then the error probability of the same/different-entity test on two families with k pooled observations decays as $\exp(-k C(P_e, P_{e'}))$ (Chernoff, 1952); in particular, the probability that a true merge—split early under weak evidence—remains undetected decays exponentially in corroborating evidence, independent of the initial segmentation.*

The assumptions are idealizations (features correlate across windows; same-name-different-entity pairs are exactly low- C cases), and the measured quantities are two engine generations, not a controlled evidence sweep. What the mechanism correspondence supports: window-batch adjudication pools a window’s candidates into one decision, and the scope-end sweep re-judges duplicate groups with full-library context to a fixed point—each raises evidence per decision. The measured joint effect is consolidation quality and ingest efficiency improving together: same-name duplicate families fall 340→124 (MAB) and 482→99 (LME) while ingest calls per document fall 46–65% (Appendix A.1). Relation-endpoint co-occurrence, injected as alignment evidence, is a high-discrimination feature in exactly this sense—shared relational neighborhoods separate the same/different hypotheses—and the same edges later widen the retrieval graph.

Store: an index, not a reconstruction. The derived layer is never answer text: all three rungs’ answerers consume verbatim spans (retrieval resolves candidates back to source episodes and snippets). Reconstruction distortion is thus structurally zero—reconstruction never occurs—and all loss is *addressing* loss: the derived layer may lose information about how to find a fact, never about the fact itself. Candidate generation runs over the family index (full-text search, embedding similarity, graph neighborhoods, reciprocal-rank fusion), so macro concept retrieval and span-level resolution form one deterministic chain whose cost scales with index size, not source bytes.

Read time: one-sided error, query-chosen distortion. Because the evidence gate admits only identifiers surfaced by tool calls and cited spans are functions of the immutable sources alone, any overlay perturbation (a stale relation, a wrong scope materialization) enters the answer only through what is read—it can change coverage, never the authenticity or auditability of what is cited. This is the formal content of “a missed scope, never a fabricated fact” (provenance reachability, not entailment, Section 3). Likewise, a summarizing memory fixes a rate-distortion point at write time that every future query inherits (Shannon, 1959); a source-grounded system defers that choice to query time—the answerer’s reading policy selects its own operating point. The ladder’s measured points—(25.1k tokens, 0.324), (35.0k, 0.407), (96.0k, 0.651) Overall, replicated across generations (Section 4.2)—trace agents choosing progressively lower-distortion points at higher rate, with the hardest verification tasks separating most (FC-MH 2–4 vs. 67–70). The framing claims this association, not single-

Benchmark	Protocol	Score	Roles
LME-S, 500 questions	sampled	84.60	answerer kimi-k3, cross-judge Qwen3.7- plus
LME full ingest, baseline	paired $n=25$	$0.68 \rightarrow 0.72$	kimi-k3
LME full ingest, tool agent	paired $n=25$	$0.68 \rightarrow 0.72$	kimi-k3
LME full ingest, sandbox	paired $n=27$	$0.889 \rightarrow 0.926$	kimi-k3 (self-judged)
LoCoMo-1540, binary J	Mem0's exact protocol	95.19	kimi-k3 answerer + judge, single run
same answers, cross-judge	re-judged	94.22 (agr. 98.51%)	judge Qwen3.7-plus

Table 6: Protocol-labeled context on other benchmarks ($v1 \rightarrow v2$ engine pairs; roles disclosed per row). The LoCoMo run differs from Mem0’s published J in two admitted ways—single run versus 10-run mean $\pm$ std, and the answerer/judge model—and carries a disclosed legacy-fingerprint caveat; it is context, not a superiority claim. The LME sandbox track’s two extra questions follow its session-evidence eligibility rule; cross-track comparisons are same-track only.

Table 7: LoCoMo per-category J (binary CORRECT/WRONG accuracy, LLM-judged). Published rows reproduced verbatim from Chhikara et al. (2025) (arXiv:2504.19413, Table 1): Mem0 reports mean $\pm$ std over 10 runs with GPT-4o-mini as both answerer and judge (std omitted, all ≤ 0.75 ; appendix). **Deep-Dream** applies the *same* J methodology with Kimi-K3 substituted for GPT-4o-mini as both answerer and judge (single run); the primary row uses the same-model judge as Mem0; the $\dagger$ row judges the same K3 answers with an independent Qwen3.7-plus cross-judge (cross-judge agreement 98.51%). Same 841/282/96/321 question populations (1,540 total, Mem0-compatible subset). *Caveat*: the K3-answerer run is flagged `formal_claim_allowed=false`—1,529 historical answer rows predate runtime-code fingerprinting (answers unchanged, sha 76af5772; per-item JSONL carries `runtime_code_sha256`); reported as a diagnostic cross-judge result, not a fingerprint-complete formal result. Deterministic F1/BLEU-1 omitted for space (Table 11). Weighted J : Mem0 ≈ 62.14 , $Mem0^g \approx 61.36$. Deep-Dream agg. J : 95.19% / 94.22% ($\dagger$).

Method	J (S/M/O/T)
LoCoMo	— / — / — / —
ReadAgent	— / — / — / —
MemoryBank	— / — / — / —
MemGPT	— / — / — / —
A-Mem	— / — / — / —
A-Mem*	39.79 / 18.85 / 54.05 / 49.91
LangMem	62.23 / 47.92 / 71.12 / 23.43
Zep	61.70 / 41.35 / 76.60 / 49.31
OpenAI	63.79 / 42.92 / 62.29 / 21.71
Mem0	67.13 / 51.15 / 72.93 / 55.51
$Mem0^g$	65.71 / 47.19 / 75.71 / 58.13
Deep-Dream	97.38 / 93.62 / 80.21 / 95.33
Deep-Dream†	96.91 / 92.91 / 77.08 / 93.46

factor causality, and the X7 equivalence bound still caps the derived channels’ offline addressing contribution at ± 2.4 points: their measured value is selection, not recall (Section 4.5).

Table 8: Published Deep Memory Retrieval (DMR) baselines reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956). DMR has 500 multi-session conversations (5 sessions, ≤ 12 messages each) with one single-hop fact-retrieval question per conversation. Deep-Dream does not currently run DMR, so this table is a contextual reference only.

Method	Answerer	Accuracy
Recursive Summarization	GPT-4-turbo	35.3%
Conversation Summaries	GPT-4-turbo	78.6%
MemGPT	GPT-4-turbo	93.4%
Full-conversation	GPT-4-turbo	94.4%
Zep	GPT-4-turbo	94.8%
Conversation Summaries	GPT-4o-mini	88.0%
Full-conversation	GPT-4o-mini	98.0%
Zep	GPT-4o-mini	98.2%

Table 9: LongMemEval main results. Published rows (GPT-4o-mini / GPT-4o, Full-context vs. Zep) reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956). The **Deep-Dream** row uses Qwen3.7-plus as both answerer and cross-judge—a different protocol; its accuracy (84.6%) is placed beside the published numbers, not as a shared leaderboard. Latency/avg-context for Deep-Dream are not measured under the published protocol (—). The six-category breakdown is in Table 10.

Answerer	System	Accuracy	Latency (s)	Avg. context
GPT-4o-mini	Full-context	55.4%	31.30	115k
GPT-4o-mini	Zep	63.8%	3.20	1.6k
GPT-4o	Full-context	60.2%	28.90	115k
GPT-4o	Zep	71.2%	2.58	1.6k
Deep-Dream	Qwen3.7-plus	84.6%	—	—

Table 10: LongMemEval six-category breakdown (binary accuracy, %). Published Full-context and Zep values reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956); the **Deep-Dream (Q.)** row is Qwen3.7-plus answerer+cross-judge on the same six categories. Different answerer/judge, so this is a matched-granularity contextual reference, not a shared leaderboard. Sess. pref. is the hardest category for all systems.

System	Sess. pref.	Sess. asst.	Temporal	Multi-sess.	Knowledge upd.	Sess. user
Full-ctx (4o-mini)	30.0	81.8	36.5	40.6	76.9	81.4
Zep (4o-mini)	53.3	75.0	54.1	47.4	74.4	92.9
Full-ctx (4o)	20.0	94.6	45.1	44.3	78.2	81.4
Zep (4o)	56.7	80.4	62.4	57.9	83.3	92.9
Deep-Dream (Q.)	53.3	100.0	84.2	74.4	91.0	98.6

Table 11: Mem0 LoCoMo deterministic metrics, reproduced verbatim from Chhikara et al. (2025) (Table 1). F1 / BLEU-1 over the four categories (Single-hop / Multi-hop / Open-domain / Temporal). Deep-Dream does not compute token-overlap metrics (its metric is binary LLM-judge accuracy, Table 7), so no Deep-Dream column appears here; this table is included for completeness of the published reproduction only.

Method	F1 (S/M/O/T)	BLEU-1 (S/M/O/T)
LoCoMo	25.02 / 12.04 / 40.36 / 18.41	19.75 / 11.16 / 29.05 / 14.77
ReadAgent	9.15 / 5.31 / 9.67 / 12.60	6.48 / 5.12 / 7.66 / 8.87
MemoryBank	5.00 / 5.56 / 6.61 / 9.68	4.77 / 5.94 / 5.16 / 6.99
MemGPT	26.65 / 9.15 / 41.04 / 25.52	17.72 / 7.44 / 34.34 / 19.44
A-Mem	27.02 / 12.14 / 44.65 / 45.85	20.09 / 12.00 / 37.06 / 36.67
A-Mem*	20.76 / 9.22 / 33.34 / 35.40	14.90 / 8.81 / 27.58 / 31.08
LangMem	35.51 / 26.04 / 40.91 / 30.75	26.86 / 22.32 / 33.63 / 25.84
Zep	35.74 / 19.37 / 49.56 / 42.00	23.30 / 14.82 / 38.92 / 34.53
OpenAI	34.30 / 20.09 / 39.31 / 14.04	23.72 / 15.42 / 31.16 / 11.25
Mem0	38.72 / 28.64 / 47.65 / 48.93	27.13 / 21.58 / 38.72 / 40.51
Mem0 ^g	38.09 / 24.32 / 49.27 / 51.55	26.03 / 18.82 / 40.30 / 40.28